CSE563 — Unit III Numerical Worksheet

Probability and Statistics

40 questions • CO3 • Solutions Booklet

Standard normal table extract

z	0.5	1.0	1.28	1.5	1.645	1.96	2.0	2.58
$\Phi(z)$	0.6915	0.8413	0.9000	0.9332	0.9500	0.9750	0.9772	0.9950

$t_{0.025,\,3} = 3.182$, $t_{0.025,\,8} = 2.306$, $t_{0.025,\,9} = 2.262$, $t_{0.025,\,15} = 2.131$, $t_{0.025,\,24} = 2.064$

$e^{-1} = 0.3679$, $e^{-1.5} = 0.2231$, $e^{-2} = 0.1353$, $e^{-3} = 0.0498$, $e^{-4} = 0.0183$

Instructor Note

Full worked solutions for all 40 questions. Conclusions to hypothesis tests are stated in context, and standard checks (posteriors summing to 1, residuals summing to zero, CI agreeing with the test) are shown.

A. Basic and Compound Probability (Q1–Q10)

Q1. Two fair dice are rolled. Find the probability that (a) the sum is 8, (b) the sum is at least 10, (c) both dice show the same value.

Solution. Sample space has 36 equally likely outcomes. **(a)** Sum 8: $(2,6),(3,5),(4,4),(5,3),(6,2)$ — 5 ways. $P = 5/36 = \mathbf{0.1389}$. **(b)** Sum $\geq$ 10: sum 10 (3 ways), 11 (2), 12 (1) = 6. $P = 6/36 = \mathbf{0.1667}$. **(c)** Doubles: 6 ways. $P = 6/36 = \mathbf{0.1667}$.

Q2. A 4-character password uses letters A–Z with repetition allowed. Find the probability that a randomly generated password (a) has all distinct characters, (b) contains at least one repeated character.

Solution. Total $= 26^4 = 456976$. **(a)** All distinct $= 26 \times 25 \times 24 \times 23 = 358800$. $P = 358800/456976 = \mathbf{0.7852}$. **(b)** Complement: $1 - 0.7852 = \mathbf{0.2148}$.

Q3. From a standard 52-card deck, 3 cards are drawn without replacement. Find the probability that all three are hearts.

Solution.

$$P = \frac{\binom{13}{3}}{\binom{52}{3}} = \frac{286}{22100} = \mathbf{0.0129}.$$

Equivalent sequential form: $\frac{13}{52} \times \frac{12}{51} \times \frac{11}{50} = 0.0129$ ✓.

Q4. A committee of 5 is selected from 6 men and 4 women. Find the probability it contains exactly 3 women.

Solution. Total $= \binom{10}{5} = 252$. Favourable $= \binom{4}{3}\binom{6}{2} = 4 \times 15 = 60$.

$$P = \frac{60}{252} = \mathbf{0.2381}.$$

Q5. $P(A) = 0.6$, $P(B) = 0.5$, $P(A \cup B) = 0.8$. Find $P(A \cap B)$, $P(A \mid B)$ and determine whether A and B are independent.

Solution. $P(A \cap B) = 0.6 + 0.5 - 0.8 = \mathbf{0.3}$. $P(A \mid B) = 0.3/0.5 = \mathbf{0.6}$. **Independent?** $P(A)P(B) = 0.6 \times 0.5 = 0.30 = P(A \cap B)$ ✓— **yes, independent**. (Consistently, $P(A \mid B) = 0.6 = P(A)$: knowing B tells us nothing about A.)

Q6. A bag has 5 red and 7 blue balls. Two are drawn without replacement. Find the probability that (a) both are red, (b) they are of different colours.

Solution. **(a)** $\frac{5}{12} \times \frac{4}{11} = \frac{20}{132} = \mathbf{0.1515}$. **(b)** $P(\text{RB}) + P(\text{BR}) = \frac{5}{12} \cdot \frac{7}{11} + \frac{7}{12} \cdot \frac{5}{11} = \frac{35+35}{132} = \frac{70}{132} = \mathbf{0.5303}$. *Check:* both blue $= \frac{7}{12} \cdot \frac{6}{11} = \frac{42}{132}$; $20 + 70 + 42 = 132$ ✓.

Q7. A system has three independent components each working with probability 0.9. Find the probability that (a) all three work, (b) at least one fails.

Solution. **(a)** $0.9^3 = \mathbf{0.729}$. **(b)** $1 - 0.729 = \mathbf{0.271}$.

Q8. A disease affects 2% of a population. A test has sensitivity 95% and specificity 90%. Find the probability that a person who tests positive actually has the disease.

Solution. $P(D) = 0.02$, $P(+ \mid D) = 0.95$, $P(+ \mid \overline{D}) = 1 - 0.90 = 0.10$.

$$P(+) = 0.02(0.95) + 0.98(0.10) = 0.019 + 0.098 = 0.117.$$

$$P(D \mid +) = \frac{0.019}{0.117} = \mathbf{0.1624}.$$

Only about 16% — the healthy group is 49 times larger and generates most of the positives. This is the **base-rate fallacy**.

Q9. Factories A, B and C supply 25%, 35% and 40% of components, with defect rates 2%, 1% and 3%. A component is found defective. Find the probability it came from each factory, and verify the posteriors sum to 1.

Solution.

$$P(D) = 0.25(0.02) + 0.35(0.01) + 0.40(0.03) = 0.0050 + 0.0035 + 0.0120 = 0.0205.$$

$$P(A \mid D) = \frac{0.0050}{0.0205} = \mathbf{0.2439}, \qquad P(B \mid D) = \frac{0.0035}{0.0205} = \mathbf{0.1707},$$

$$P(C \mid D) = \frac{0.0120}{0.0205} = \mathbf{0.5854}.$$

Check: $0.2439 + 0.1707 + 0.5854 = 1.0000$ ✓. Factory C is the most likely source — it has both the largest share and the highest defect rate.

Q10. Two machines produce 60% and 40% of items, with 3% and 5% defective respectively. Find (a) the overall defect rate, (b) the probability a defective item came from machine 2.

Solution. **(a)** $P(D) = 0.6(0.03) + 0.4(0.05) = 0.018 + 0.020 = \mathbf{0.038}$ (3.8%). **(b)** $P(M_2 \mid D) = \dfrac{0.020}{0.038} = \mathbf{0.5263}$. Machine 2 makes only 40% of output but over half the defects.

B. Probability Distributions (Q11–Q20)

Q11. A fair coin is tossed 8 times. Find the probability of exactly 5 heads, and the mean and variance of the number of heads.

Solution. $X \sim \text{Bin}(8, 0.5)$.

$$P(X = 5) = \binom{8}{5}(0.5)^8 = \frac{56}{256} = \mathbf{0.21875}.$$

Mean $= 8(0.5) = \mathbf{4}$; variance $= 8(0.5)(0.5) = \mathbf{2}$; $\sigma = 1.414$.

Q12. 20% of items from a production line are defective. In a sample of 10, find the probability of (a) exactly 3 defective, (b) no defectives, (c) at least 2 defective.

Solution. $X \sim \text{Bin}(10, 0.2)$. **(a)** $\binom{10}{3}(0.2)^3(0.8)^7 = 120(0.008)(0.2097) = \mathbf{0.2013}$. **(b)** $(0.8)^{10} = \mathbf{0.1074}$. **(c)** $P(X \geq 2) = 1 - P(0) - P(1) = 1 - 0.1074 - 10(0.2)(0.8)^9 = 1 - 0.1074 - 10(0.2)(0.1342) = 1 - 0.1074 - 0.2684 = \mathbf{0.6242}$.

Q13. A factory finds that 1% of its chips are faulty. In a batch of 300, use the Poisson approximation to find the probability of at most 2 faulty chips.

Solution. n large, p small, so Poisson with $\lambda = np = 300(0.01) = 3$.

$$P(X \leq 2) = e^{-3}\left(1 + 3 + \frac{9}{2}\right) = 0.0498(8.5) = \mathbf{0.4232}.$$

Q14. Emails arrive at a server at 4 per hour. Find the probability of (a) exactly 2 in an hour, (b) none in 30 minutes, (c) at least 1 in 15 minutes.

Solution. **(a)** $\lambda = 4$: $P(2) = \dfrac{e^{-4}4^2}{2!} = \dfrac{0.0183(16)}{2} = \mathbf{0.1465}$. **(b)** 30 min $\Rightarrow \lambda = 2$: $P(0) = e^{-2} = \mathbf{0.1353}$. **(c)** 15 min $\Rightarrow \lambda = 1$: $P(X \geq 1) = 1 - e^{-1} = 1 - 0.3679 = \mathbf{0.6321}$. *Note the rescaling of λ in each part — this is the commonest source of error.*

Q15. Defects occur in a cable at 1.5 per kilometre. Find the probability that a 2 km stretch has exactly 3 defects.

Solution. For 2 km, $\lambda = 3$.

$$P(X=3) = \frac{e^{-3}3^3}{3!} = \frac{0.0498(27)}{6} = \mathbf{0.2241}.$$

Q16. Heights are normally distributed with $\mu = 165$ cm and $\sigma = 10$ cm. Find (a) $P(X > 175)$, (b) $P(150 < X < 180)$, (c) the height exceeded by only 5% of people.

Solution. **(a)** $z = 1.0$: $P = 1 - 0.8413 = \mathbf{0.1587}$. **(b)** $z_1 = -1.5$, $z_2 = 1.5$: $0.9332 - 0.0668 = \mathbf{0.8664}$. **(c)** $z = 1.645$: $x = 165 + 1.645(10) = \mathbf{181.45}$ cm.

Q17. Component lifetimes are $N(\mu = 1200, \sigma = 100)$ hours. Find the probability a component lasts (a) more than 1400 hours, (b) between 1100 and 1300 hours. (c) Find the 90th percentile.

Solution. **(a)** $z = 2.0$: $1 - 0.9772 = \mathbf{0.0228}$. **(b)** $z = \pm 1.0$: $0.8413 - 0.1587 = \mathbf{0.6826}$ (the 68% rule). **(c)** $z = 1.28$: $x = 1200 + 128 = \mathbf{1328}$ hours.

Q18. A random variable X has $P(X{=}1) = 0.2$, $P(X{=}2) = 0.5$, $P(X{=}3) = 0.3$. Find $E[X]$, $\mathrm{Var}(X)$ and $E[3X - 1]$.

Solution. $E[X] = 1(0.2) + 2(0.5) + 3(0.3) = 0.2 + 1.0 + 0.9 = \mathbf{2.1}$. $E[X^2] = 1(0.2) + 4(0.5) + 9(0.3) = 0.2 + 2.0 + 2.7 = 4.9$. $\mathrm{Var}(X) = 4.9 - 2.1^2 = 4.9 - 4.41 = \mathbf{0.49}$; $\sigma = 0.7$. $E[3X - 1] = 3(2.1) - 1 = \mathbf{5.3}$.

Q19. In a game, you win Rs.50 with probability 0.2, win Rs.10 with probability 0.3 and lose Rs.20 with probability 0.5. Find the expected gain and state whether the game is fair.

Solution.

$$E[X] = 50(0.2) + 10(0.3) + (-20)(0.5) = 10 + 3 - 10 = \mathbf{Rs.3}.$$

The expected gain is positive (Rs.3 per play), so the game is **not fair** — it favours the player. A fair game has $E[X] = 0$.

Q20. The number of daily server crashes follows a Poisson distribution with mean 0.5. Find the probability of (a) no crashes in a day, (b) more than 1 crash in a day, (c) no crashes in a 7-day week. *Use* $e^{-0.5} = 0.6065$, $e^{-3.5} = 0.0302$.

Solution. **(a)** $P(0) = e^{-0.5} = \mathbf{0.6065}$. **(b)** $P(X > 1) = 1 - e^{-0.5}(1 + 0.5) = 1 - 1.5(0.6065) = 1 - 0.9098 = \mathbf{0.0902}$. **(c)** Week: $\lambda = 3.5$, so $P(0) = e^{-3.5} = \mathbf{0.0302}$.

C. Frequentist Inference (Q21–Q30)

Q21. A sample of $n = 100$ has $\bar{x} = 52$ and known $\sigma = 10$. Construct 95% and 99% confidence intervals for μ.

Solution. SE $= 10/10 = 1$. **95%:** $52 \pm 1.96(1) = [\mathbf{50.04}, \mathbf{53.96}]$. **99%:** $52 \pm 2.58(1) = [\mathbf{49.42}, \mathbf{54.58}]$. Higher confidence gives a wider interval — more certainty costs precision.

Q22. A sample of 25 light bulbs has $\bar{x} = 1050$ hours and $s = 100$ hours. Construct a 95% confidence interval for the mean lifetime.

Solution. σ unknown and n small, so use t with 24 df: $t_{0.025,24} = 2.064$. SE $= 100/5 = 20$.

$$1050 \pm 2.064(20) = 1050 \pm 41.28 = [\mathbf{1008.72},\ \mathbf{1091.28}] \text{ hours.}$$

Q23. In a poll, 420 of 1000 voters support a proposal. Construct a 95% confidence interval for the true proportion.

Solution. $\hat{p} = 0.42$, SE $= \sqrt{\dfrac{0.42(0.58)}{1000}} = \sqrt{0.0002436} = 0.01561$.

$$0.42 \pm 1.96(0.01561) = 0.42 \pm 0.0306 = [\mathbf{0.3894},\ \mathbf{0.4506}].$$

Since the interval lies entirely below 0.5, there is evidence the proposal lacks majority support.

Q24. How large a sample is needed to estimate a mean to within ± 2 units at 95% confidence, given $\sigma = 15$?

Solution. Require $1.96\dfrac{15}{\sqrt{n}} \le 2$:

$$\sqrt{n} \ge \frac{1.96(15)}{2} = 14.7 \Rightarrow n \ge 216.09.$$

$n = 217$ (always round *up*).

Q25. A manufacturer claims a mean weight of 250 g. A sample of $n = 36$ gives $\bar{x} = 244$ g with known $\sigma = 12$ g. Test at $\alpha = 0.05$ (two-sided).

Solution. $H_0 : \mu = 250$ vs $H_1 : \mu \neq 250$. SE $= 12/6 = 2$.

$$z = \frac{244 - 250}{2} = \mathbf{-3.0}.$$

Critical values ± 1.96; $|-3.0| > 1.96$, so **reject** H_0. p-value $= 2(1 - \Phi(3.0)) = 2(1 - 0.9987) = 0.0026 < 0.05$ ✓. **Conclusion:** there is strong evidence at the 5% level that the true mean weight differs from the claimed 250 g, and the sample suggests it is lower.

Q26. A sample of 9 measurements gives $\bar{x} = 10.4$ and $s = 0.6$. Test $H_0 : \mu = 10$ against $H_1 : \mu \neq 10$ at $\alpha = 0.05$.

Solution. t-test with 8 df; $t_{0.025,8} = 2.306$. SE $= 0.6/3 = 0.2$.

$$t = \frac{10.4 - 10}{0.2} = \mathbf{2.0}.$$

$|2.0| < 2.306$, so **fail to reject** H_0. **Conclusion:** the data do not provide sufficient evidence at the 5% level that the mean differs from 10. *CI check:* $10.4 \pm 2.306(0.2) = [9.94, 10.86]$, which contains 10 ✓.

Q27. A new algorithm is claimed to reduce mean runtime below 200 ms. A sample of $n = 49$ gives $\bar{x} = 195$ ms, $\sigma = 21$ ms. Test at $\alpha = 0.05$ (one-sided).

Solution. $H_0 : \mu = 200$ vs $H_1 : \mu < 200$ (one-sided, lower tail). SE $= 21/7 = 3$.

$$z = \frac{195 - 200}{3} = \mathbf{-1.667}.$$

Critical value for a lower-tailed test at 5%: -1.645. Since $-1.667 < -1.645$, **reject** H_0. p-value $= \Phi(-1.667) \approx 0.0478 < 0.05$ ✓. **Conclusion:** there is evidence at the 5% level that the new algorithm's mean runtime is below 200 ms. Note the result is marginal — a two-sided test would *not* have rejected.

Q28. For a test of $H_0 : \mu = 50$ with $\sigma = 10$, $n = 25$ and $\alpha = 0.05$ (upper-tailed), find the rejection boundary in original units, and compute β and the power when the true mean is 55.

Solution. SE $= 10/5 = 2$. Rejection boundary: $\bar{x} > 50 + 1.645(2) = \mathbf{53.29}$.
Under $\mu = 55$:

$$\beta = P(\bar{X} \le 53.29) = P\left(Z \le \frac{53.29 - 55}{2}\right) = P(Z \le -0.855) \approx \mathbf{0.196}.$$

Power $= 1 - 0.196 = \mathbf{0.804}$ — the test detects a 5-unit shift about 80% of the time.

Q29. Explain, with the relevant probabilities, the difference between a Type I and a Type II error for a test conducted at $\alpha = 0.01$ with power 0.85.

Solution. **Type I error** — rejecting H_0 when it is true (a false alarm). Its probability is $\alpha = \mathbf{0.01}$, i.e. 1% of the time we would declare an effect that does not exist.
Type II error — failing to reject H_0 when H_1 is true (a miss). Since power $= 1 - \beta = 0.85$, we have $\beta = \mathbf{0.15}$: 15% of the time a real effect of the stated size would go undetected.
The two trade off at fixed n: lowering α to 0.01 (a strict test) raises β. Only increasing the sample size reduces both simultaneously.

Q30. A 95% confidence interval for a mean is $[23.5,\ 28.5]$. State the sample mean and margin of error, and state whether $H_0 : \mu = 30$ would be rejected at $\alpha = 0.05$.

Solution. $\bar{x} = \dfrac{23.5 + 28.5}{2} = \mathbf{26}$; margin of error $= \mathbf{2.5}$.
Since 30 lies *outside* the interval, $H_0 : \mu = 30$ **would be rejected** at the 5% level in a two-sided test. A two-sided test at level α rejects exactly those null values excluded by the $100(1-\alpha)$% confidence interval.

D. Bayesian Inference (Q31–Q34)

Q31. A coin is flipped 15 times giving 11 heads. With a uniform Beta$(1, 1)$ prior, find the posterior distribution, its mean, and the MAP estimate. Compare with the MLE.

Solution. Posterior $=$ Beta$(1 + 11,\ 1 + 4) =$ Beta$(\mathbf{12}, \mathbf{5})$.

$$\text{Mean} = \frac{12}{17} = \mathbf{0.7059}, \qquad \text{MAP} = \frac{11}{15} = \mathbf{0.7333}, \qquad \text{MLE} = \frac{11}{15} = \mathbf{0.7333}.$$

With a uniform prior the MAP coincides with the MLE; the posterior *mean* is slightly lower because the Beta$(12, 5)$ density is skewed.

Q32. Repeat the previous question with an informative Beta$(10, 10)$ prior. Explain the difference.

Solution. Posterior $=$ Beta$(10 + 11,\ 10 + 4) =$ Beta$(\mathbf{21}, \mathbf{14})$.

$$\text{Mean} = \frac{21}{35} = \mathbf{0.600}, \qquad \text{MAP} = \frac{20}{33} = \mathbf{0.606}.$$

Difference: the Beta$(10, 10)$ prior encodes a fairly firm belief that the coin is fair, worth 20 pseudo-observations. Against only 15 real flips it pulls the estimate from 0.733 down to 0.600 — substantial *shrinkage* toward 0.5. With 1500 flips at the same rate the prior would be negligible.

Q33. A quality inspector uses a Beta$(2, 8)$ prior for the defect rate. A batch of 50 items contains 8 defectives. Find the posterior mean and compare it with the sample proportion.

Solution. Posterior $=$ Beta$(2 + 8,\ 8 + 42) =$ Beta$(10, 50)$.

$$\text{Posterior mean} = \frac{10}{60} = \mathbf{0.1667}.$$

Sample proportion (MLE) $= 8/50 = 0.16$; prior mean $= 2/10 = 0.20$. The posterior 0.1667 lies between the two, much closer to the data because 50 observations outweigh the prior's 10 pseudo-counts.

Q34. A naive Bayes spam filter has $P(\text{spam}) = 0.4$. The word "win" occurs in 40% of spam and 2% of ham; "invoice" in 5% of spam and 25% of ham. Classify a message containing both words.

Solution.

$$\text{score}(\text{spam}) = 0.4 \times 0.40 \times 0.05 = 0.0080,$$

$$\text{score}(\text{ham}) = 0.6 \times 0.02 \times 0.25 = 0.0030.$$

$$P(\text{spam} \mid \mathbf{x}) = \frac{0.0080}{0.0080 + 0.0030} = \frac{0.0080}{0.0110} = \mathbf{0.7273}.$$

Classified as spam (probability 0.727). The strong "win" signal outweighs the ham-leaning "invoice".

E. Regression Analysis (Q35–Q40)

Q35. For $x = 1, 3, 5, 7, 9$ and $y = 3, 7, 10, 16, 19$, compute $\bar{x}$, $\bar{y}$, S_{xx}, S_{yy} and S_{xy}.

Solution. $\bar{x} = 25/5 = \mathbf{5}$; $\bar{y} = 55/5 = \mathbf{11}$.

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$	$(y - \bar{y})^2$	product
1	3	-4	-8	16	64	32
3	7	-2	-4	4	16	8
5	10	0	-1	0	1	0
7	16	2	5	4	25	10
9	19	4	8	16	64	32
				$S_{xx} = \mathbf{40}$	$S_{yy} = \mathbf{170}$	$S_{xy} = \mathbf{82}$

Q36. Using the previous question's values, find the least-squares regression line and interpret the slope.

Solution.

$$\hat{\beta}_1 = \frac{82}{40} = \mathbf{2.05}, \qquad \hat{\beta}_0 = 11 - 2.05(5) = 11 - 10.25 = \mathbf{0.75}.$$

$$\boxed{\hat{y} = 0.75 + 2.05x}$$

Interpretation: each one-unit increase in x is *associated with* an increase of about 2.05 units in y. The intercept 0.75 is the fitted value at $x = 0$, which lies outside the observed range and should not be over-interpreted. *Check:* the line passes through $(\bar{x}, \bar{y})$: $0.75 + 2.05(5) = 11$ ✓.

Q37. Compute r and R^2 for the data of Q35, and state what R^2 means here.

Solution.

$$r = \frac{82}{\sqrt{40 \times 170}} = \frac{82}{\sqrt{6800}} = \frac{82}{82.462} = \mathbf{0.9944}.$$

$$R^2 = r^2 = \mathbf{0.9888}.$$

Meaning: 98.9% of the variation in y across these observations is explained by the linear relationship with x; only 1.1% remains as unexplained residual variation.

Q38. For the fitted line $\hat{y} = 0.75 + 2.05x$, compute the fitted values and residuals for all five points, and verify the residuals sum to zero. Hence find SSE.

Solution.

x	y	$\hat{y}$	$e = y - \hat{y}$
1	3	2.80	+0.20
3	7	6.90	+0.10
5	10	11.00	−1.00
7	16	15.10	+0.90
9	19	19.20	−0.20
		$\sum e =$	**0.00** ✓

$$\text{SSE} = 0.04 + 0.01 + 1.00 + 0.81 + 0.04 = \mathbf{1.90}.$$

Check against R^2: $1 - 1.90/170 = 1 - 0.01118 = 0.9888$ ✓— matches r^2.

Q39. Test whether the slope of the regression in Q36 is significantly different from zero at $\alpha = 0.05$, given $t_{0.025,3} = 3.182$.

Solution.

$$\text{SE}(\hat{\beta}_1) = \sqrt{\frac{\text{SSE}/(n-2)}{S_{xx}}} = \sqrt{\frac{1.90/3}{40}} = \sqrt{\frac{0.6333}{40}} = \sqrt{0.015833} = 0.1258.$$

$$t = \frac{2.05}{0.1258} = \mathbf{16.29} \quad \text{on 3 df.}$$

Since $16.29 > 3.182$, **reject** $H_0 : \beta_1 = 0$. **Conclusion:** there is strong evidence that x is linearly related to y; the slope is significantly non-zero at the 5% level.

Q40. Using $\hat{y} = 0.75 + 2.05x$, predict y at $x = 4$ and at $x = 25$. Comment on the reliability of each prediction.

Solution. At $x = 4$: $\hat{y} = 0.75 + 8.20 = \mathbf{8.95}$. At $x = 25$: $\hat{y} = 0.75 + 51.25 = \mathbf{52.00}$.
Reliability. The prediction at $x = 4$ is *interpolation* — 4 lies inside the observed range $[1, 9]$ where the linear fit was validated ($R^2 = 0.989$), so it is well supported.
The prediction at $x = 25$ is **extrapolation**, nearly three times beyond the largest observed x. Nothing in the data justifies assuming the linear relationship continues that far; the true relationship may curve, saturate or break down entirely. This prediction should be treated as unreliable and reported with that caveat.